

WEB MONITOR — POTENTIAL FUTURE WORK

Research directions following the 2004 UIUC master's thesis, the historical Perl reconstruction, and the 2026 Python cloud deployment. These are possible extensions and research questions, not outstanding implementation commitments.

1. Domain-aware extraction and change detection

Move beyond URL membership and normalized page-text hashes toward identifying meaningful domain-specific changes, such as newly published research papers or updated course materials, while filtering irrelevant changes and supporting dynamically rendered websites.

2. Adaptive crawling and scheduling

Investigate focused crawling and change-frequency prediction to prioritize relevant pages and dynamically adjust monitoring intervals, rather than crawling every configured website on a fixed daily schedule.

3. Semantic and cross-source monitoring

Extend monitoring beyond individual web pages by applying information-integration techniques, including mediated schemas, semantic mappings, and entity matching, to identify meaningful changes across multiple related sources. This revisits a research direction explicitly proposed in the 2004 thesis.

4. Scalability, consistency, and fault tolerance

Evaluate alternative persistence and execution architectures that provide stronger transactional guarantees, concurrency control, and recovery from partial failures. The current ActivityInfo-backed implementation deliberately favors simplicity, serialized execution, and low-cost deployment over distributed scalability.

5. Evaluating AI-assisted software modernization

Use the documented Perl reconstruction and Python reimplementations as a case study for investigating how AI coding assistants support legacy-system comprehension, migration, testing, and maintenance. Future research could examine developer productivity, correctness, human oversight, and the reproducibility of AI-assisted engineering decisions.

SOURCE CONTEXT

2004 thesis, Chapter 4, especially §§4.5–4.7:

<https://github.com/sobulo/sobulo-web-monitor-2004/blob/main/source/SobuloMasterThesis.pdf>

Historical reconstruction audit:

<https://github.com/sobulo/sobulo-web-monitor-2004/blob/main/docs/static-audit.md>

Modern architecture and tradeoffs:

<https://github.com/sobulo/web-monitor/blob/main/docs/architecture.md>

STATUS

Documentation-only retrospective. The completed implementations are not being reopened;